

Multiple Choice (20 marks)

1. What drives differentiation?
 - (a) Spontaneous emission from radioactive atoms.
 - (b) The minimization of gravitational potential energy.
 - (c) Thermally induced collisions.
 - (d) Plate tectonics.
2. Why do we see essentially the same face of the Moon at all times?
 - (a) because the Moon has a nearly circular orbit around the Earth
 - (b) because the Moon does not rotate
 - (c) because the other face points toward us only at new moon when we can't see the Moon
 - (d) because the Moon's rotational and orbital periods are equal
3. Previous IAAC rounds featured Proxima/Alpha Centauri as closest star(system) to the Earth. Which one is the second closest star(system)?
 - (a) Wolf 359
 - (b) Sirius
 - (c) 61 Cygni
 - (d) Barnard's Star
4. Why is the sky blue?
 - (a) Because the molecules that compose the Earth's atmosphere have a blue-ish color.
 - (b) Because the sky reflects the color of the Earth's oceans.
 - (c) Because the atmosphere preferentially scatters short wavelengths.
 - (d) Because the Earth's atmosphere preferentially absorbs all other colors.
5. Which of the following characteristics would not necessarily suggest that a rock we found is a meteorite.
 - (a) It has a fusion crust
 - (b) It contains solidified spherical droplets
 - (c) It is highly processed
 - (d) It has different elemental composition than earth
6. The sky is blue because

- (a) the Sun mainly emits blue light.
 - (b) the atmosphere absorbs mostly blue light.
 - (c) molecules scatter red light more effectively than blue light.
 - (d) molecules scatter blue light more effectively than red light.
7. Approximately how old is the surface of Venus?
- (a) 750 million years.
 - (b) 2 billion years.
 - (c) 3 billion years.
 - (d) 4.5 billion years.
8. Why doesn't Venus have seasons like Mars and Earth do?
- (a) Its rotation axis is nearly perpendicular to the plane of the Solar System.
 - (b) It does not have an ozone layer.
 - (c) It does not rotate fast enough.
 - (d) It is too close to the Sun.
9. What is the so-called bolometric luminosity in astronomy?
- (a) The luminosity integrated over vertically polarized wavelengths.
 - (b) The luminosity integrated over horizontally wavelengths.
 - (c) The luminosity integrated over visible wavelengths.
 - (d) The luminosity integrated over all wavelengths.
10. Which is not a similarity between Saturn and Jupiter's atmospheres?
- (a) a composition dominated by hydrogen and helium
 - (b) the presence of belts zones and storms
 - (c) an equatorial wind speed of more than 900 miles per hour
 - (d) significant "shear" between bands of circulation at different latitudes

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'From shortest to longest wavelength which of the following correctly orders the different categories of electromagnetic radiation?' is 'gamma rays X rays ultraviolet visible light infrared radio'.
2. True or False: The correct answer to 'Which is not an essential requirement for life as we know it?' is 'A source of organic molecules'.

3. True or False: The correct answer to 'Why do we look for water-ice in craters at Mercury's pole?' is 'These craters contain the only permanently shadowed regions on Mercury'.
4. True or False: The correct answer to 'Which of these has NOT been one of the main hypotheses considered for the origin of the Moon?' is 'The Moon split from the Earth due to tidal forces.'.
5. True or False: The correct answer to 'The distance between the Earth and the star Altair is one million times greater than the distance between the Earth and the Sun. How far...' is ' 1.5×10^{14} meters'.
6. True or False: The correct answer to 'When was the telescope invented by Galileo?' is '1609'.
7. True or False: The correct answer to 'What does the astronomical term ecliptic describe?' is 'The path of the Sun in the sky throughout a year.'.
8. True or False: The correct answer to 'Sunspots are black regions that temporarily appear on the Sun. Their number highly increases every ... years. This period is also called...' is '11'.
9. True or False: The correct answer to 'We were first able to accurately measure the diameter of Pluto from:' is 'brightness measurements made during mutual eclipses of Pluto and Charon'.
10. True or False: The correct answer to 'What is true for a type-Ia ("type one-a") supernova?' is 'This type produces gamma-ray bursts.'.

End of Paper